COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
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ACTIVITY NO. 7.1 **Managing Files and Creating Roles in Ansible**

1. PROCEDURE OUTPUT

Task 1: Create a file and copy it to remote servers

mkdir files, nvim default_site.html

```
<!DOCTYPE html>
<html lang="en">
<head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>My Web Server</title>
</head>
<body>
        <h1>Welcome to My Web Server</h1>

        <p>This webpage was deployed using Ansible.</p>
        <p>This file was copied from the local repository to the remote server.</p>

</body>
</html>
```

run ansible-playbook –ask-become-pass site.yaml

```
PLAY RECAP *********************************************************************************
centos1                    : ok=5    changed=1    unreachable=0    failed=0    skipped=2
rescued=0    ignored=0
ubuntu1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=1
rescued=0    ignored=0
ubuntu2                    : ok=5    changed=1    unreachable=0    failed=0    skipped=3
rescued=0    ignored=0
```

- It runs but the ubuntu1 failed to install update. The silly reason is that I forgot to continue the update the server T_T. but rest assured it works now after “fixing the problem”.
- Now the site.yaml made sure that the newly created default_site.html is working through ansible-playbook.

```
PLAY RECAP *********************************************************************************
centos1                    : ok=5    changed=1    unreachable=0    failed=0    skipped=2
rescued=0    ignored=0
ubuntu1                    : ok=5    changed=1    unreachable=0    failed=0    skipped=3
rescued=0    ignored=0
ubuntu2                    : ok=5    changed=0    unreachable=0    failed=0    skipped=3
rescued=0    ignored=0
```

ssh ubuntu1

```
ubuntu1@server1:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>My Web Server</title>
</head>
<body>
        <h1>Welcome to My Web Server</h1>

        <p>This webpage was deployed using Ansible.</p>
        <p>This file was copied from the local repository to the remote server.</p>

</body>
</html>
ubuntu1@server1:~$ _
```

ssh ubuntu2

```
ubuntu2@server2:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>My Web Server</title>
</head>
<body>
        <h1>Welcome to My Web Server</h1>

        <p>This webpage was deployed using Ansible.</p>
        <p>This file was copied from the local repository to the remote server.</p>

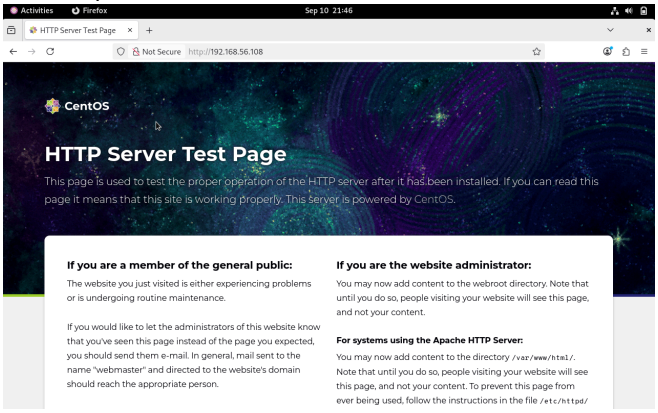
</body>
</html>
ubuntu2@server2:~$
```

ssh centos1

```
[centos1@localhost ~]$ cat /var/www/html/index.html
cat: /var/www/html/index.html: No such file or directory
[centos1@localhost ~]$
```

- Both control node prints the same output after running site.yaml except centos

centos1 ip add on browser



Syncing repository

```
localserver@workstation:~/CPE232_Duenas$ git add .
localserver@workstation:~/CPE232_Duenas$ git commit -m "HOA 7.1 task1"
[main b10f31b] HOA 7.1 task1
 2 files changed, 26 insertions(+)
 create mode 100644 ansible-lab/files/default_site.html
localserver@workstation:~/CPE232_Duenas$ git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 871 bytes | 871.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To github.com:mo-na-min/CPE232_Duenas.git
   a1c967e..b10f31b  main -> main
localserver@workstation:~/CPE232_Duenas$ git pull
Already up to date.
```

Task 2: Download a file and extract it to a remote server

Edit site.yaml

```
- hosts: workstations
  become: true

  tasks:
    - name: install unzip
      package:
        name: unzip

    - name: install terraform
      unarchive:
        src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.
zip
        dest: /usr/local/bin
        remote_src: yes
        mode: 0775
        ownder: root
        group: root
```

Edit inventory.yaml

```
workstations:
  hosts:
    ubuntu1:
```

Running the playbook

```
PLAY RECAP *********************************************************************************
centos1                    : ok=5    changed=1    unreachable=0    failed=0    skipped=2
rescued=0    ignored=0
ubuntu1                    : ok=8    changed=1    unreachable=0    failed=0    skipped=3
rescued=0    ignored=0
ubuntu2                    : ok=5    changed=0    unreachable=0    failed=0    skipped=3
rescued=0    ignored=0
```

- What happened is that the execution of the workstations play on the designated host (ubuntu1). It first runs the install unzip. Next, the install terraform task uses the unarchive module to download the Terraform zip archive directly from hachicorp's URL and extracts it to /usr/local/bin with root privileges and 0755 permissions. All in all, no failures.

Verifying installation of "terraform" in ubuntu1

```
ubuntu1@server1:~$ terraform
Usage: terraform [-version] [-help] <command> [args]

The available commands for execution are listed below.
The most common, useful commands are shown first, followed by
less common or more advanced commands. If you're just getting
started with Terraform, stick with the common commands. For the
other commands, please read the help and docs before usage.

Common commands:
    apply              Builds or changes infrastructure
    console            Interactive console for Terraform interpolations
    destroy            Destroy Terraform-managed infrastructure
    env                Workspace management
    fmt                Rewrites config files to canonical format
    get                Download and install modules for the configuration
    graph              Create a visual graph of Terraform resources
    import             Import existing infrastructure into Terraform
    init               Initialize a Terraform working directory
    login              Obtain and save credentials for a remote host
    logout             Remove locally-stored credentials for a remote host
    output             Read an output from a state file
    plan               Generate and show an execution plan
    providers          Prints a tree of the providers used in the configuration
    refresh            Update local state file against real resources
    show               Inspect Terraform state or plan
    taint              Manually mark a resource for recreation
    untaint            Manually unmark a resource as tainted
    validate           Validates the Terraform files
    version            Prints the Terraform version
    workspace          Workspace management

All other commands:
    0.12upgrade        Rewrites pre-0.12 module source code for v0.12
    debug              Debug output management (experimental)
    force-unlock       Manually unlock the terraform state
    push               Obsolete command for Terraform Enterprise legacy (v1)
    state              Advanced state management
ubuntu1@server1:~$
```

Task 3: Create roles

Edit site.yaml

```
localserver@workstation:~/CPE232_Duenas/ansible-lab$ cp site.yaml oldsite.yaml
localserver@workstation:~/CPE232_Duenas/ansible-lab$ ls
ansible.cfg  install_apache.yaml  oldsite.yaml   roles.yaml  testvars.yaml
files        inventory.yaml       playbook.yaml  site.yaml
```

```
---

- hosts: all
  become: true
  pre_tasks:

    - name: update repository index (CentOS)
      tags: always
      dnf:
        update_only: yes
      changed_when: false
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      tags: always
      apt:
        update_cache: yes
        changed_when: false
      when: ansible_distribution == "Ubuntu"
```

```
- hosts: all
  become: true
  roles:
    - base

- hosts: workstations
  become: true
  roles:
    - workstations

- hosts: web_servers
  become: true
  roles:
    - web_servers

- hosts: db_servers
  become: true
  roles:
    -db_servers

- hosts: file_servers
  become: true
  roles:
    - file_servers
```

Make a directory “roles”.

```
localserver@workstation:~/CPE232_Duenas/ansible-lab$ mkdir roles
localserver@workstation:~/CPE232_Duenas/ansible-lab$ ls
ansible.cfg  install_apache.yaml  oldsite.yaml   roles       site.yaml
files        inventory.yaml       playbook.yaml  roles.yaml  testvars.yaml
localserver@workstation:~/CPE232_Duenas/ansible-lab$ cd roles/
localserver@workstation:~/CPE232_Duenas/ansible-lab/roles$ mkdir base web_servers file_serve
rs db_servers workstations
localserver@workstation:~/CPE232_Duenas/ansible-lab/roles$ ls
base  db_servers  file_servers  web_servers  workstations
```

```
localserver@workstation:~/CPE232_Duenas/ansible-lab/roles$ for d in ~/CPE232_Duenas/ansible-lab/roles/*/tasks; do cp ~/CPE232_Duenas/ansible-lab/oldsite.yaml "$d/main.yml"; done
localserver@workstation:~/CPE232_Duenas/ansible-lab/roles$ tree
.
├── base
│   └── tasks
│       └── main.yml
├── db_servers
│   └── tasks
│       └── main.yml
├── file_servers
│   └── tasks
│       └── main.yml
├── web_servers
│   └── tasks
│       └── main.yml
└── workstations
    └── tasks
        └── main.yml
```

Running the site.yaml

```
localserver@workstation:~/CPE232_Duenas/ansible-lab$ ansible-playbook --ask-become-pass site.yaml
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [base : install updates (Ubuntu)] *****************************************************
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

TASK [base : install updates (CentOS)] *****************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY [db_servers] **************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [centos1]

TASK [db_servers : install mariadb package (CentOS)] ***************************************
ok: [centos1]

TASK [db_servers : Mariadb - Restarting/Enabling] ******************************************
changed: [centos1]

TASK [db_servers : install mariadb package (Ubuntu)] ***************************************
skipping: [centos1]
[WARNING]: Could not match supplied host pattern, ignoring: file_servers

PLAY [file_servers] ************************************************************************
skipping: no hosts matched

PLAY [web_servers] *************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [ubuntu1]
ok: [ubuntu2]
```

```
TASK [web_servers : install apache and php for Ubuntu servers] *****************************
ok: [ubuntu2]
ok: [ubuntu1]

TASK [web_servers : install apache and php for CentOS servers] *****************************
skipping: [ubuntu1]
skipping: [ubuntu2]

TASK [web_servers : start httpd (CentOS)] **************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]

TASK [web_servers : copy default html file for site] ***************************************
ok: [ubuntu2]
ok: [ubuntu1]

PLAY [workstations] ************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [ubuntu1]

TASK [workstations : install unzip] ********************************************************
ok: [ubuntu1]

TASK [workstations : install terraform] ****************************************************
ok: [ubuntu1]

PLAY RECAP *********************************************************************************
centos1                    : ok=5    changed=1    unreachable=0    failed=0    skipped=2
rescued=0    ignored=0
ubuntu1                    : ok=8    changed=0    unreachable=0    failed=0    skipped=3
rescued=0    ignored=0
ubuntu2                    : ok=5    changed=0    unreachable=0    failed=0    skipped=3
rescued=0    ignored=0

localserver@workstation:~/CPE232_Duenas/ansible-lab$
```

- I successfully created roles for each tasks.

Synchronizing repository

```
localserver@workstation:~/CPE232_Duenas$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   ansible-lab/inventory.yaml
        modified:   ansible-lab/site.yaml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        ansible-lab/oldsite.yaml
        ansible-lab/roles/

no changes added to commit (use "git add" and/or "git commit -a")
```

```
localserver@workstation:~/CPE232_Duenas$ git add .
localserver@workstation:~/CPE232_Duenas$ git commit -m "HOA 7.1 files"
[main 4bd8ae4] HOA 7.1 files
 8 files changed, 221 insertions(+), 83 deletions(-)
 create mode 100644 ansible-lab/oldsite.yaml
 create mode 100644 ansible-lab/roles/base/tasks/main.yml
 create mode 100644 ansible-lab/roles/db_servers/tasks/main.yml
 create mode 100644 ansible-lab/roles/file_servers/tasks/main.yml
 create mode 100644 ansible-lab/roles/web_servers/tasks/main.yml
 create mode 100644 ansible-lab/roles/workstations/tasks/main.yml
localserver@workstation:~/CPE232_Duenas$ git push origin main
Enumerating objects: 26, done.
Counting objects: 100% (26/26), done.
Delta compression using up to 2 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (22/22), 2.69 KiB | 1.35 MiB/s, done.
Total 22 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To github.com:mo-na-min/CPE232_Duenas.git
   b10f31b..4bd8ae4  main -> main
localserver@workstation:~/CPE232_Duenas$ git pull
Already up to date.
localserver@workstation:~/CPE232_Duenas$
```

Github link: https://github.com/mo-na-min/CPE232_Duenas

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of creating roles?
   - Modularizing playbooks into organized, reusable folder structures via making separate directories.
2. What is the importance of managing files?
   - Keeps configuration files and web content identical across all remote nodes.

Conclusion:

I worked in a lab where I used Ansible file modules. I used them to put web pages in place and to copy scripts to different nodes. I tested this on Ubuntu and also on CentOS. I also rebuilt a large playbook into smaller Ansible roles. That made the setup easier to reuse. I then added checks for the operating system. With those checks, the same tasks could run on many types of nodes without me changing everything each time.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of

concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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